



ISP Environmental Solutions Pvt. Ltd.
MANUFACTURER • SMART RECYCLING SYSTEMS

Public RVM Operator & User Manual

Safe daily operation, public recycling workflow, rewards, screen guidance and first-response troubleshooting

Manufactured by: ISP Environmental Solutions Pvt. Ltd.

Product family: ISP Public Reverse Vending Machine (RVM)

Application: RVMDesktopApp

Environment: Shopping malls, transit stations, commercial hubs and municipal parks

Materials: Plastic bottles, beverage cans, UBC/Tetra Pak cartons and rejected items

Edition: 1 October 2026

This manual intentionally contains only normal operating controls. Confidential service and administrative controls are excluded.

1. Purpose and safety

Manufactured by ISP Environmental Solutions Pvt. Ltd. • Smart Recycling Systems

The public RVM accepts eligible empty containers, validates them automatically, updates the live session breakdown and awards points through the supported mobile/reward flow.

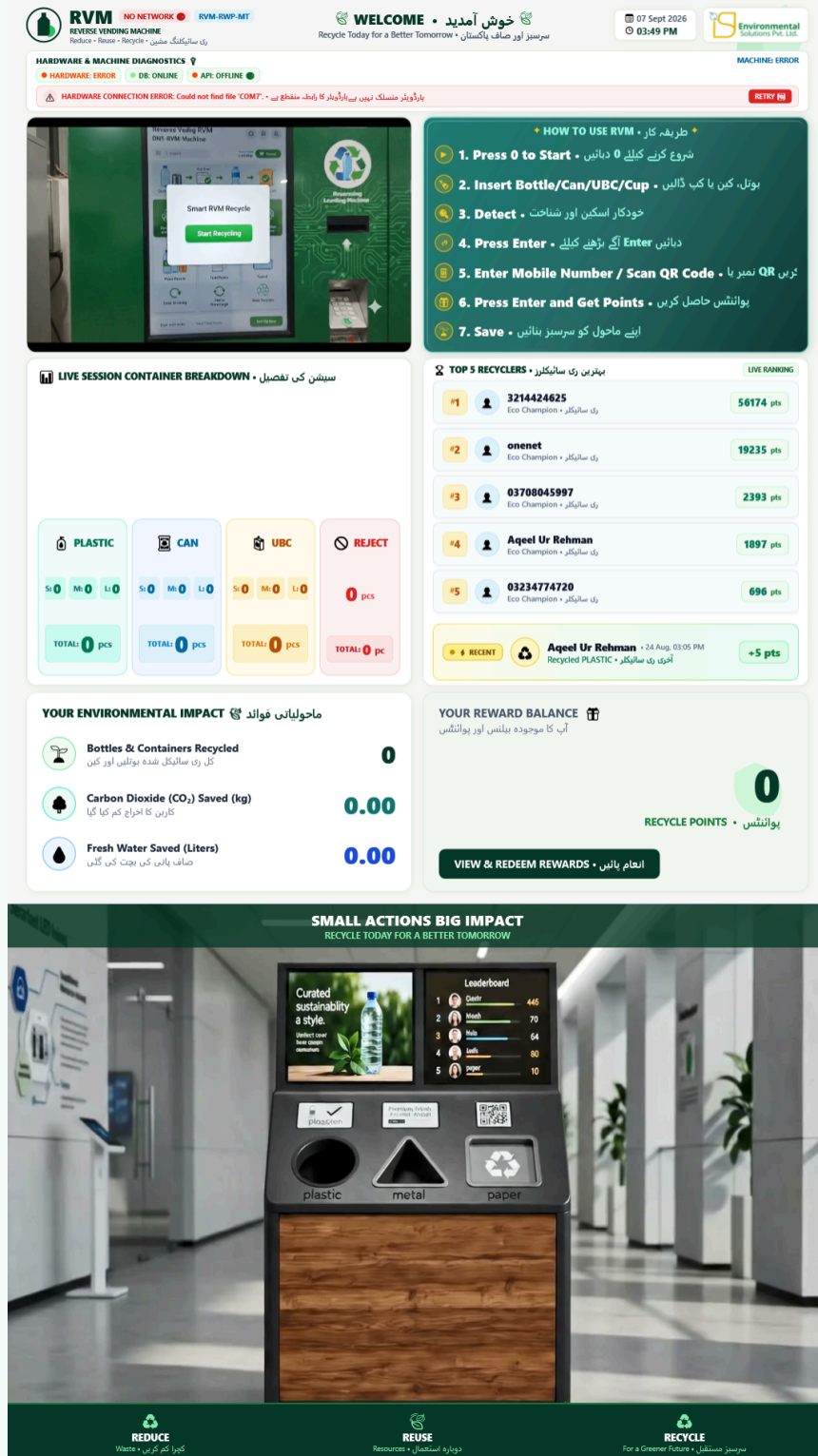
Safety: Never place hands inside the intake, force an item through a gate, insert liquids, glass, batteries, pressurized containers, sharp objects or general waste. If the machine reports a blockage or unavailable compartment, stop public use and contact the assigned operator.

People and responsibilities

Role	Normal responsibility
Public user	Follow screen prompts, insert one eligible item at a time, claim rewards and remove rejected items.
Site operator	Inspect cleanliness and safety, confirm normal status, replace full-bin liners using approved procedures and report faults.
Technical support	Electrical, sensor, gate, serial, software and database maintenance. Not covered by this user manual.

2. Know the main screen

Manufactured by ISP Environmental Solutions Pvt. Ltd. • Smart Recycling Systems



Screen capture: RVM welcome dashboard. The diagnostic banner, instructions, session breakdown, leaderboard, environmental impact and reward balance are visible.

- **Status area:** Shows machine, hardware, database and API availability.
- **Instruction area:** Displays the current action and supporting video.
- **Live breakdown:** Counts plastic, cans, UBC and rejects by the configured size groups.
- **Reward balance:** Shows session points.
- **Leaderboard:** Displays current public rankings.

3. Accepted and prohibited items

Manufactured by ISP Environmental Solutions Pvt. Ltd. • Smart Recycling Systems

Category	Generally accepted	Do not insert
Plastic	Empty PET/HDPE bottles in supported sizes; cap and label rules follow the screen/site policy.	Liquid-filled, chemical, oily, heavily crushed or unrecognizable plastic.
Can	Empty beverage cans in supported sizes.	Aerosols, batteries, food tins with residue, sharp or heavily crushed metal.
UBC	Empty beverage cartons/Tetra Pak in supported sizes.	Wet, leaking, contaminated or flattened material the sensors cannot identify.
Reject	Items that fail material, size, safety or transit validation are returned/not credited.	Never reinsert repeatedly if the same clean eligible item is rejected.

Preparation: Empty the container, remove straws and loose waste, keep its basic shape, and insert only one item when instructed.

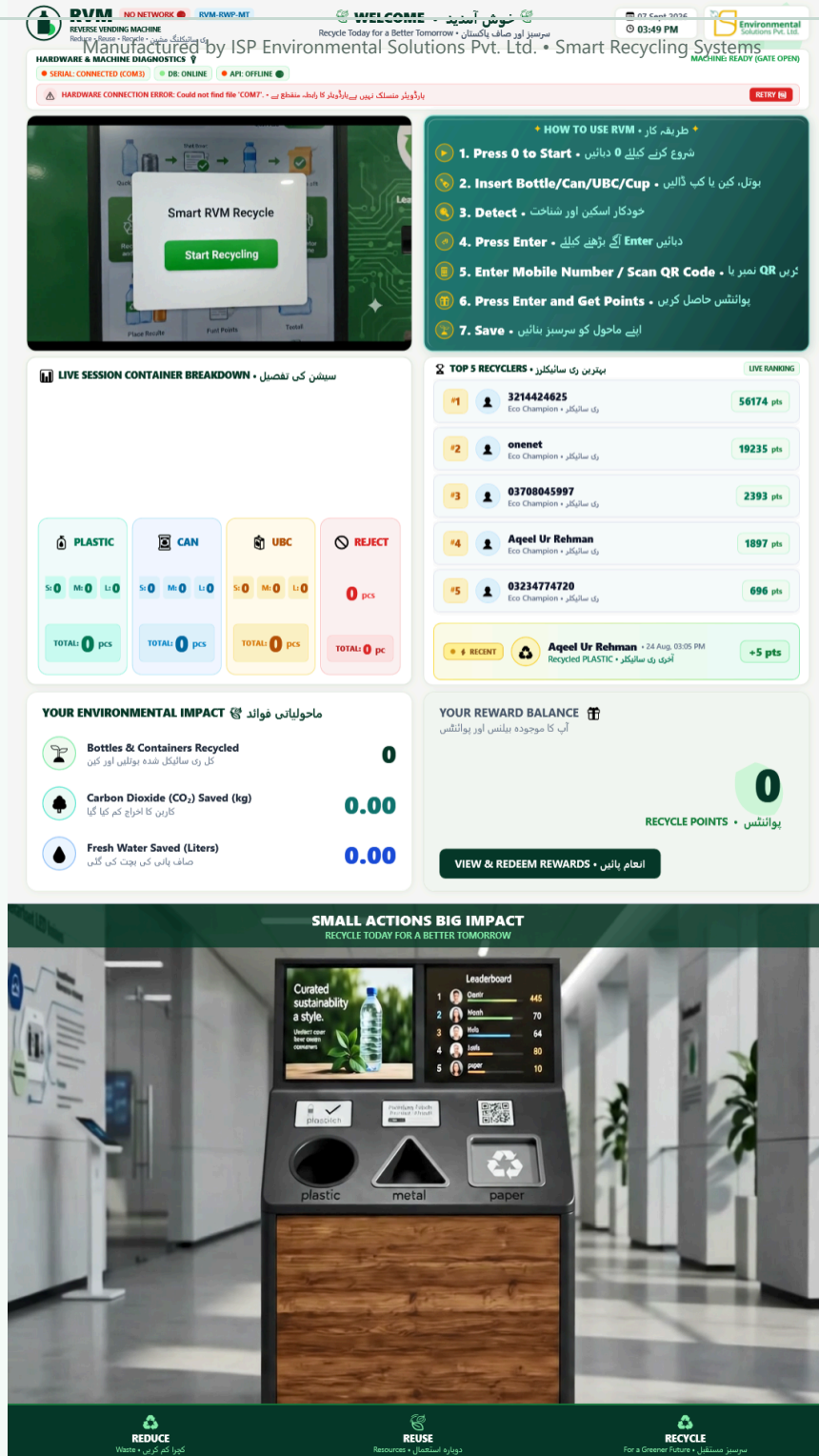
4. Start a recycling session

Touchless start

Use the mobile application to scan the on-screen start QR code when available. Wait for the kiosk to confirm the session.

Keypad start

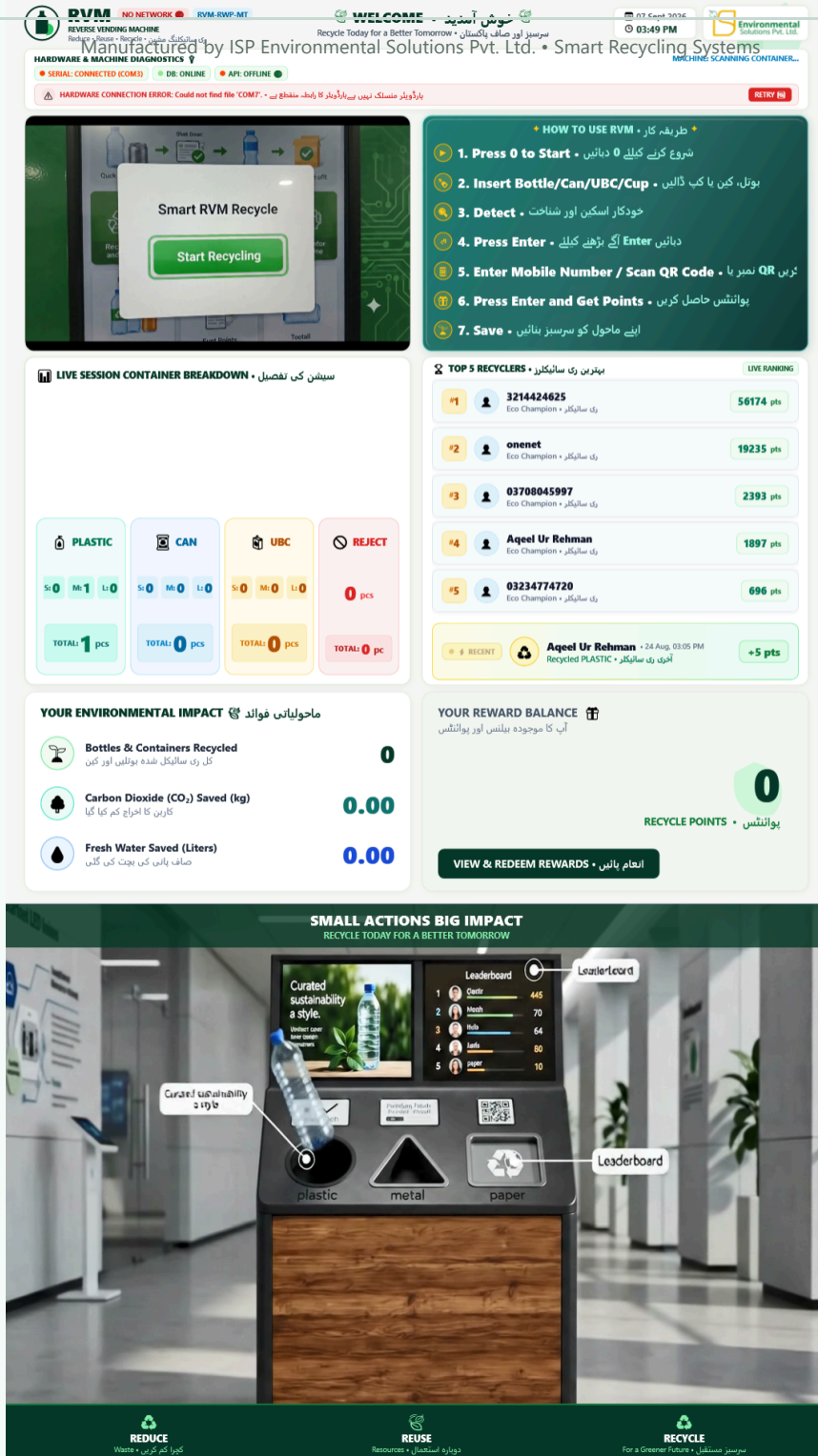
Press operational key  once. Wait for “Machine Started” or the insert instruction before approaching the intake.



Screen capture: active session waiting for one container.

5. Insert and validate items

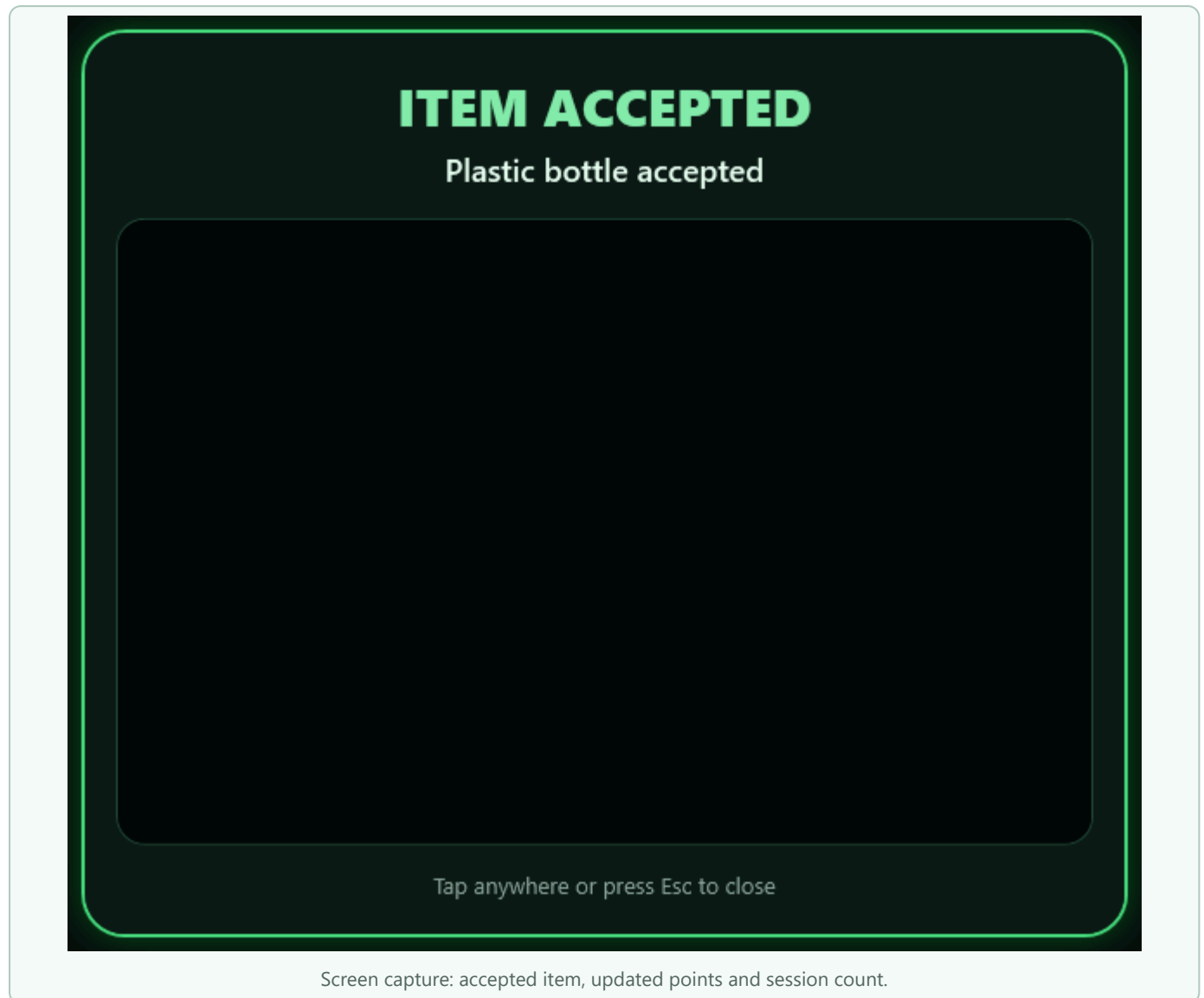
1. Confirm the machine is ready and the desired material is available.
2. Insert one clean, empty container in the orientation demonstrated on screen.
3. Release it; do not hold, rotate or push it during scanning.
4. Wait while material and size sensors validate the item.
5. Read the accepted or rejected result before inserting the next item.



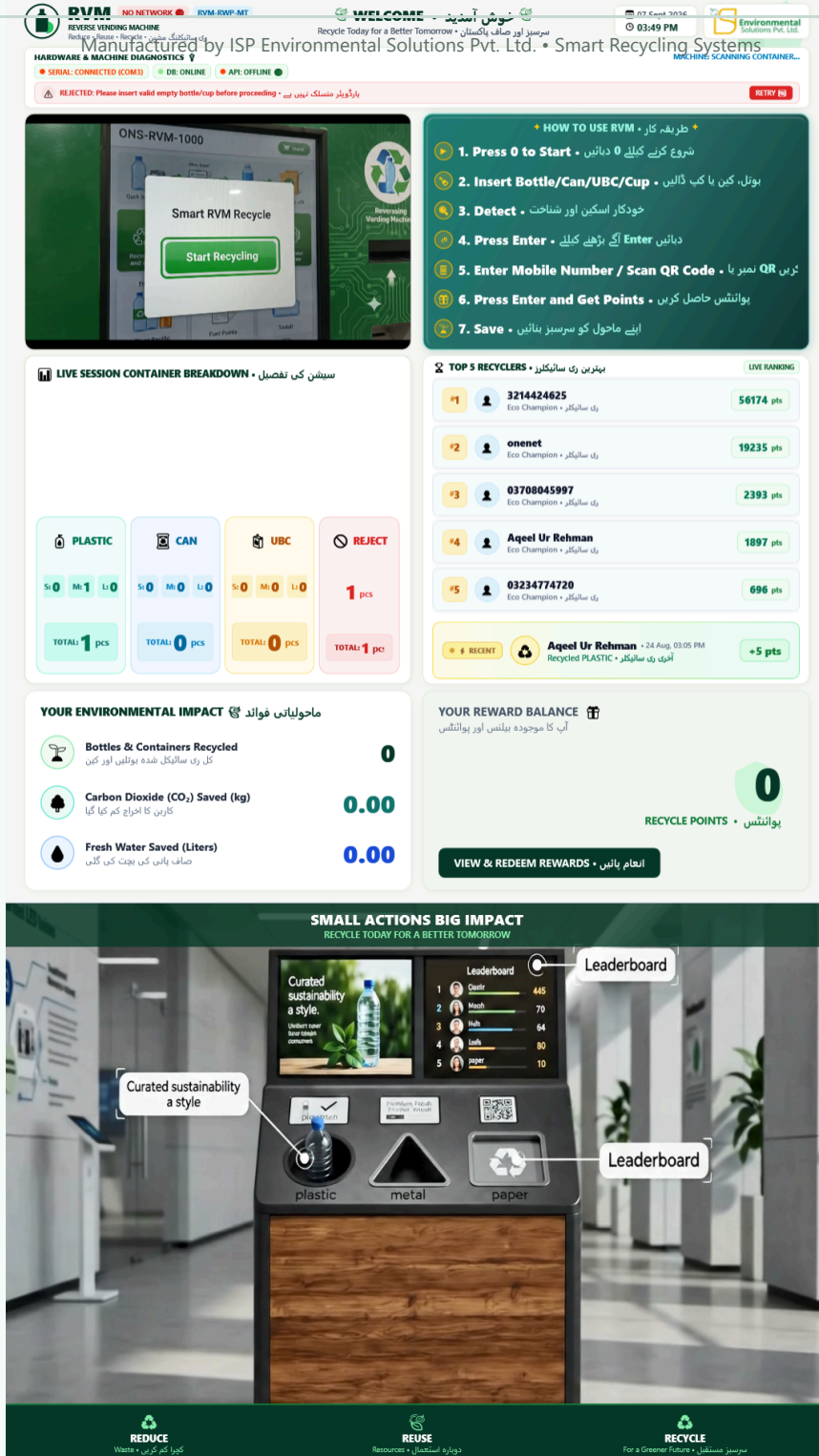
Screen capture: detection and sizing in progress. Do not insert another item.

6. Accepted and rejected outcomes

Manufactured by ISP Environmental Solutions Pvt. Ltd. • Smart Recycling Systems



After acceptance, confirm the live material count and reward total increased. Insert another item only after the ready prompt returns.



Screen capture: rejection result.

For rejection, allow all movement to stop, remove the returned item if instructed, check that it is empty and eligible, and retry only once. Repeated rejection should be reported to the site operator.

7. Finish and claim rewards

1. When all items are processed, press operational key **Enter**.
2. Scan the displayed session QR using the supported mobile app, or enter the requested mobile number with the numeric keypad.

3. Review the number before confirming with **Enter**.
4. Wait for the success confirmation; do not walk away while the transaction is processing.

WALLET TRANSFER · Send Points to Wallet

RECYCLED DEPOSIT
3 Item(s) Recycled

POINTS TO CREDIT
+30 PTS

MOBILE NUMBER (03xxxxxxxxxx)

03001234567

Must start with 03 and contain exactly 11 digits (e.g. 03001234567).

✓ Valid mobile number

X Cancel Credit Wallet ✓

Screen capture: reward-claim/mobile-number stage.

Privacy: Enter contact information only on the official claim screen. Shield the keypad and never provide passwords, OTPs or wallet PINs to the kiosk or another person.

8. Rating and completion

If feedback is requested, select the displayed 1–5 rating using the touchscreen or numeric keypad and follow the confirmation prompt. Wait until the completion message appears and the kiosk returns to welcome mode.

9. Operator opening checklist

- ☐ Cabinet, screen, intake and return area are clean and undamaged.
- ☐ No hands, objects or liquid are inside the intake.
- ☐ Screen is full-screen and responsive.
- ☐ Machine/hardware status is ready; no persistent red fault banner.
- ☐ Correct machine identity and location are displayed.
- ☐ Network/API status is normal or approved offline mode is understood.
- ☐ All intended material compartments show available.
- ☐ Receipt/reward flow is available.
- ☐ One approved test item completes successfully where site procedure permits.

10. First-response troubleshooting

Message/symptom	Safe response
-----------------	---------------

Hardware offline/unplugged	Stop public deposits. Check only externally accessible power/cable indicators. Notify technical support; do not open energized panels. Marsippos Ltd. • Smart Recycling Systems
Material unavailable/bin full	Block that material intake and arrange approved bin service. Never bypass a full-bin warning.
Item rejected	Remove it when safe, verify eligibility and retry once. Use another machine or contact the operator if it repeats.
Item stuck or gate does not complete	Do not reach inside or force the gate. Isolate the machine and call technical support.
Reward not received	Record machine ID, approximate time, mobile/account identifier and displayed points. Do not repeat the deposit solely to generate another credit.
Screen frozen	Keep users away from the intake and escalate through the approved support channel.

11. Operator closing checklist

- ☐ Allow active users to complete their sessions.
- ☐ Confirm no item remains in the intake/return area.
- ☐ Record unresolved rejects, jams, reward cases and warning banners.
- ☐ Clean exterior surfaces with approved materials; never spray liquid into openings.
- ☐ Follow the site’s approved shutdown or always-on policy.

Escalate immediately: smoke, burning smell, exposed wiring, liquid ingress, unusual mechanical noise, repeated gate motion, damaged glass, vandalism or any risk to the public.